

|       |                                                                |           |
|-------|----------------------------------------------------------------|-----------|
|       |                                                                |           |
| 10(a) | single straight line along full length of solenoid             | <b>B1</b> |
|       | at least two more parallel lines along full length of solenoid | <b>B1</b> |
|       | correct direction – right to left                              | <b>B1</b> |
| 10(b) | (induced) e.m.f. proportional / equal to <u>rate</u>           | <b>M1</b> |
|       | of change of (magnetic) flux (linkage)                         | <b>A1</b> |
| 10(c) | increasing current causes increasing flux                      | <b>B1</b> |
|       | increasing flux induces e.m.f. in coil                         | <b>B1</b> |
|       | (induced) e.m.f. opposes growth of current                     | <b>B1</b> |